

JEE MAIN REVISION MASTER: UNITS & MEASUREMENTS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 227 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dimensional Analysis & Principle of Homogeneity	125 PYQs	Dimensions
Very High	Error Analysis & Significant Figures	123 PYQs	Errors

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dimensional Analysis & Principle of Homogeneity	$[G] = M^{-1}L^3T^{-2}$, $[h] = ML^2T^{-1}$, $[\mu_0] = MLT^{-2}A^{-2}$
Error Analysis & Significant Figures	$\frac{\Delta Z}{Z} = a\frac{\Delta A}{A} + b\frac{\Delta B}{B} + c\frac{\Delta C}{C}$ $\text{for } Z = \frac{A^a B^b}{C^c}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dimensional Analysis & Principle of Homogeneity (Very High Yield — 125 PYQs)

Core Principle: Application of dimensions to check equation consistency and derive relations between physical quantities.

■ **Caution / JEE Pitfall:** Trigonometric, logarithmic, and exponential arguments must strictly be dimensionless.

■ **Speed Trick:** Equate powers of M, L, T on both sides to find unknown exponents.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T)..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "The time period of a simple harmonic oscillator is $T=2\pi\sqrt{(k/m)}$. Measured value of mass (m) of the object is 10 g with an a..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "If ϵ, E and t represent the free space permittivity, electric field and time respectively, then the unit of ($\epsilon E / t$)..." [Ans: C]

3.2 Error Analysis & Significant Figures (Very High Yield — 123 PYQs)

Core Principle: Maximum fractional and percentage errors in combination of quantities and screw gauge / vernier calipers reading.

■ **Caution / JEE Pitfall:** Errors always ADD up in worst-case analysis; never subtract fractional errors.

■ **Speed Trick:** Least count = Pitch / (Number of circular divisions). Total reading = MSR + (CSR × LC) - (Zero Error).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth..." **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "The density ρ of a uniform cylinder is determined by measuring its mass m , length l and diameter d . The measured values of $m, l, \dots$ " **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7^{th} ..." **[Ans: C]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance?

- (A) 200α
- (B) 400α **[Correct]**
- (C) 300α
- (D) 500α

Correct Answer: B

Solution Analysis: To find the distance between two points when an object moves at a constant speed, we use formula, Distance = Speed $\times$ Time α is a new unit of length such that 1 unit (α) = speed of light in vacuum (c). The speed of light c is the distance light travels per unit of time (usually meters per second). $1\alpha = c \times (1\text{second})$ The time taken for light to travel from Venus to Earth is 6 minutes and 40 seconds. 1 minute = 60 seconds 6 minutes = $6 \times 60 = 360$ seconds Total time $t = 360\text{s} + 40\text{s} = 400$ seconds Using distance formula: Distance = Speed of light $\times$ Time Distance $= c$

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#1)

Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are _____.

- (A) $[h T L M^{-2}]$
- (B) $[h T^{-1} L M^{-2}]$ **[Correct]**
- (C) $[h T L^2 M^{-2}]$
- (D) $[h^{-1} T^{-1} L M^{-2}]$

Correct Answer: B

Solution Analysis: To find the dimensions of the universal gravitational constant G in terms of Planck's constant h , length L , mass M , and time T , we must first determine the base dimensions of both G and h in the standard MLT system. According to Newton's law of universal gravitation, the force F between two masses m_1 and m_2 separated by a distance r is given by: $F = \frac{G m_1 m_2}{r^2} \Rightarrow G = \frac{F r^2}{m_1 m_2}$ Substituting the dimensions $[M L T^{-2}]$ for force, $[L]$ for distance (r), and mass (m) as $[M]$: $[G] = \frac{[M L T^{-2}] [L^2] [M] [M]}{[M]^2} = [M^{-1} L^3 T^{-2}]$